

Chapter 1

Countability

1.1 Introduction

In the introductory module (MATH40001), we constructed the number systems $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. Real analysis focuses on the real numbers, $\mathbb{R}$, formalising many properties about functions that we know from calculus. For example, an intuitive notion of a continuous function is that you can draw the function on a piece of paper without taking your pen off the page. We make concepts like these precise in this module.

1.2 Countability

Before starting this chapter (and, in general, this module), it is imperative that you are familiar with the logic and sets part of the introductory module, MATH40001.

Definition 1.1. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.

Intuitively, this says that you can list the elements of A as $a_1, a_2, \dots$ using only the natural numbers. We say that a set is *countable* if it is finite or countably infinite. Note we will take $\mathbb{N} = \{1, 2, \dots\}$, unless stated otherwise.

Note that if there is a bijection between a given pair of sets A and B , then A is countable if and only if B is countable. This extends the natural idea of finite sets having the same number of elements if and only if there is a bijection between the sets. Formally, we define two sets having the same *cardinality* as follows:

Definition 1.2. Let A, B be sets. We say that A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$.

The idea is that even in infinite settings, if we know a property about the 'size' of one set A , and there exists a bijection with another set B , then B shares this notion of 'size'. For example, as we mentioned, if such a bijection exists between two sets A, B , then A is countable if and only if B is countable. This gives rise to an equivalence relation, which we define as cardinality.

Exercise. Show that the relation $\sim$ on $\mathcal{P}(\mathcal{U})$ (where $\mathcal{U}$ is the universal set of interest) defined by $A \sim B \iff \exists f : A \rightarrow B$ such that f is a bijection, is an equivalence relation.

Theorem 1.3. *If $A \subseteq \mathbb{N}$ is an infinite set, then A is countably infinite.*

Proof. We construct a function $f : \mathbb{N} \rightarrow A$ inductively: since A is nonempty, and $A \subseteq \mathbb{N}$, by well-ordering of the natural numbers we can pick $a_1 = \min A$, and let $f(1) = a_1$. Assume that, for $k \geq 1$, $a_1, \dots, a_k$ have been chosen and $f(i) = a_i$ for $1 \leq i \leq k$. Since A is infinite, $A \setminus \{a_1, \dots, a_k\}$ is nonempty. Pick $a_{k+1} = \min(A \setminus \{a_1, \dots, a_k\})$, and let $f(k+1) = a_{k+1}$. Thus by construction f is strictly increasing. We claim that f is a bijection.

- Injective: this holds because by construction f is strictly increasing. We saw in the introductory module that strictly increasing functions are injective.
- Surjective: suppose for contradiction that f is not surjective. Then there exists $a \in A$ such that for all $n \in \mathbb{N}$, $f(n) \neq a$. Consider $A \setminus \{a\}$.

□

Lemma 1.4. *Let S be an infinite set. There exists an injection $f : \mathbb{N} \rightarrow S$.*

Proof. We construct a function $f : \mathbb{N} \rightarrow S$ inductively: since S is infinite, it is nonempty, so we can pick $s_1 \in S$ and define $f(1) := s_1$. Suppose for $k \in \mathbb{N}$, $s_1, \dots, s_k$ have been chosen and we have defined $f(i) = s_i$ for $1 \leq i \leq k$. Since S is infinite, $S \setminus \{s_1, \dots, s_k\}$ is nonempty, so there is $s_{k+1} \in S \setminus \{s_1, \dots, s_k\}$. Define $f(k+1) := s_{k+1}$. By construction, if $i < j$ are natural numbers, then $s_j \in S \setminus \{s_1, \dots, s_{j-1}\}$ so $s_j \neq s_i$. So f is an injection. □

As a result, we can always *embed* the natural numbers into any infinite set via an injection $f : \mathbb{N} \rightarrow S$ (where S is any infinite set). In this way, the natural numbers can be thought of as 'the smallest type of infinity'.

A not so surprising fact which we can prove immediately is that the set of integers, $\mathbb{Z}$, is countably infinite:

Theorem 1.5. *$\mathbb{Z}$ is countably infinite.*

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ even} \\ -\frac{(n+1)}{2}, & \text{if } n \text{ odd} \end{cases}$$

Then f is injective: suppose $f(a) = f(b)$. If a and b are of different parity then $f(a)$ and $f(b)$ are of different sign, so cannot be equal. So this case cannot happen. If a and b are even, then $a/2 = b/2$ so $a = b$. If a and b are odd, then $-(a+1)/2 = -(b+1)/2$ so $a = b$. So f is injective. Also, f is surjective: let $m \in \mathbb{Z}$ be arbitrary. If $m < 0$ then $f(-2m-1) = m$, and if $m \geq 0$ then $f(2m) = m$. Thus f is a bijection and $\mathbb{Z}$ is countable. □

Remark. This proof used a useful trick called interleaving, where we listed the integers as $0, -1, 1, -2, 2, \dots$, interleaving the negative and positive integers to create the desired bijection between $\mathbb{N}$ and $\mathbb{Z}$.

A less trivial fact is that the rational numbers, $\mathbb{Q}$, is countable. This may be hard to see until observing the proof of the following fact:

Theorem 1.6. $\mathbb{N}^2$ is countably infinite.

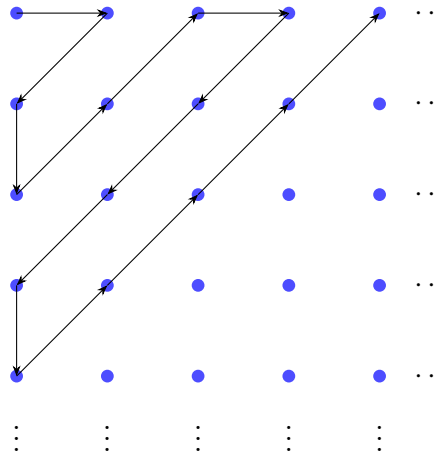


Figure 1.1: The diagonal listing of $\mathbb{N}^2$, showing it is countably infinite.

Proof. The idea is illustrated in the above diagram. Note that the listing goes

$$(0, 0), (0, 1), (1, 0), (2, 0), (1, 1), (0, 2), \dots$$

Notice that along each diagonal, $a + b$ is constant for all pairs (a, b) in that diagonal. This, along with the triangular number formula, provides the motivation for how to construct a bijection between $\mathbb{N}^2$ and $\mathbb{N}$.

Explicitly, define $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ by $f(a, b) = \frac{(a+b)(a+b+1)}{2} + b$. This is a bijection (exercise) so $\mathbb{N}^2$ is countable. Note that here we took $\mathbb{N} = \{0, 1, 2, \dots\}$. $\square$

As a consequence, we get:

Theorem 1.7. $\mathbb{Q}$ is countably infinite.

Proof. First we use Theorem 1.6 to show that $\mathbb{Q}_{\geq 0} := \{x \in \mathbb{Q} : x \geq 0\}$ is countably infinite. Let

$$S = \{(a, b) \in \mathbb{N}^2 : b \geq 1, \gcd(a, b) = 1\}.$$

Then $f : S \rightarrow \mathbb{Q}_{\geq 0}$, defined by $f(a, b) = a/b$ is a bijection because we know that all rational numbers in $\mathbb{Q}_{\geq 0}$ take the unique form a/b where $b \geq 1$ and $\gcd(a, b) = 1$. Since $\mathbb{N}^2$ is countably infinite by Theorem 1.6, $S \subseteq \mathbb{N}^2$ is also countably infinite, so $\mathbb{Q}_{>0}$ is countably infinite.

It follows (exercise) that $\mathbb{Q}$ is countably infinite, using the same idea (interleaving) as the proof that $\mathbb{Z}$ is countably infinite. $\square$

Exercise. Finish the proof of Theorem 1.7. (*Hint:* use the fact we proved above that $\mathbb{Q}_{>0}$ is countably infinite, so there exists a bijection $h : \mathbb{N} \rightarrow \mathbb{Q}_{>0}$).

The following lemma is a useful way of proving that a given set S is countably infinite without explicitly constructing a bijection $f : \mathbb{N} \rightarrow S$.

Lemma 1.8 (A criterion for countability). *Let S be an infinite set.*

1. *S is countably infinite if and only if there exists an injection $\iota : S \rightarrow \mathbb{N}$.*
2. *S is countably infinite if and only if there exists a surjection $g : \mathbb{N} \rightarrow S$.*

Proof. a □

The idea for item (1) above is that if you can *embed* your set into the natural numbers via an injection $\iota : S \rightarrow \mathbb{N}$, then it must be countably infinite, as $\iota(S) \subseteq \mathbb{N}$, and we have shown infinite subsets of $\mathbb{N}$ are countable, so since ι is an injection, S has the same cardinality as $\iota(S)$, (i.e. $S \sim \iota(S)$), so S is countably infinite. The idea for item (2) is similar: if you can use the natural numbers to list all the elements of your infinite set S , then even though this surjective map $g : \mathbb{N} \rightarrow S$ may not be injective, we have nonetheless listed all the elements of S using only the natural numbers, so S must be countably infinite also.

A simple example using Lemma 1.8 is that finite products of countable sets are countable (note that we already have the tools to prove this without Lemma 1.8 by using Theorem 1.6, which is a more elementary way of proving this fact via an explicit bijection).

Before looking at the example, try and think of properties of the integers that we can use to construct an injection $\iota : A \times B \rightarrow \mathbb{N}$.

Example 1.9. If A, B are countably infinite sets, then $A \times B$ is countably infinite.

Proof. Recall from the introductory module that we define

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

Since A and B are countably infinite, let $f : \mathbb{N} \rightarrow A$, $g : \mathbb{N} \rightarrow B$ be bijections. Therefore, we can write $A = \{a_1, a_2, \dots\}$ and $B = \{b_1, b_2, \dots\}$. Define $\iota : A \times B \rightarrow \mathbb{N}$ by $\iota(a_i, b_j) = 2^i 3^j$. By the Fundamental Theorem of Arithmetic from the introductory module, ι is an injection because prime factorisations are unique, so by Lemma 1.8, $A \times B$ is countably infinite. □

This trick of mapping each element of the set to a prime factorisation is useful to keep in mind, as it is a very neat way of proving countability for sets.

Not all sets are countably infinite:

Definition 1.10. An infinite set is called *uncountable* if it is not countably infinite.

For example, the set of real numbers $\mathbb{R}$ is uncountable. To show this, we first prove:

Lemma 1.11. *The set $(0, 1) := \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable.*

Proof. The main technique that we use in arguing why a given set is uncountable is *Cantor's diagonal argument*, which goes like this. Suppose, for contradiction, that $(0, 1)$ is countably infinite. Then there exists a bijection $f : \mathbb{N} \rightarrow (0, 1)$ (note here we will let $\mathbb{N} = \{1, 2, \dots\}$ for aesthetics; the argument does not change). For

each $n \in \mathbb{N}$, we write $f(n) = x_n = 0.a_{1n}a_{2n}\dots$, writing each x_n in decimal form, excluding tails of the digit 9 (for example, $0.8999\dots = 0.9$). We now get a listing of the numbers in $(0, 1)$ which looks like:

$$\begin{aligned} f(1) &= 0.a_{11}a_{12}a_{13}a_{14}\dots \\ f(2) &= 0.a_{21}a_{22}a_{23}a_{24}\dots \\ f(3) &= 0.a_{31}a_{32}a_{33}a_{34}\dots \\ f(4) &= 0.a_{41}a_{42}a_{43}a_{44}\dots \\ &\vdots \end{aligned}$$

The idea is that we look at the *diagonal* elements, a_{ii} for $i \geq 1$, and construct a number $x \in (0, 1)$ such that for all $i \geq 1$, the i^{th} decimal place of x differs from the i^{th} decimal place of $f(i)$, thereby constructing a number $x \in (0, 1)$ that is not in the enumeration f of the interval $(0, 1)$, which is a contradiction to f being surjective. We construct $x \in (0, 1)$ as follows: for $j \geq 1$, let b_j be the j^{th} decimal place of x , then we define

$$b_j = \begin{cases} 2 & \text{if } a_{jj} = 1 \\ 1 & \text{if } a_{jj} \neq 1 \end{cases}$$

Then by construction, $b_i \neq a_{ii}$ for each $i \geq 1$, so in particular (by uniqueness of decimal representations under our restriction to decimal expansions with no digit 9 tails), we have $x \neq f(i)$ for all $i \geq 1$, so $x \in (0, 1)$ is not in the image of f , a contradiction to f being surjective onto $(0, 1)$. Therefore, $(0, 1)$ is uncountable. $\square$

Theorem 1.12. $\mathbb{R}$ is uncountable.

Proof. Follows from Lemma 1.11 (exercise). $\square$

Exercise. Find a bijection $f : (0, 1) \rightarrow \mathbb{R}$, and deduce that $\mathbb{R}$ is an uncountable set.